

An explicit irrationality-exponent bound for $\zeta(3) - r\zeta(2)$

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Abstract

For every fixed rational number r , we prove $2 \leq \mu(\zeta(3) - r\zeta(2)) \leq 10000$. More precisely, every reduced rational approximant a/q with sufficiently large denominator satisfies $|\zeta(3) - r\zeta(2) - a/q| > q^{-10000}$; the threshold may depend on r . The proof extends Tang's determinant construction by estimating the conditioning of an associated positive Gram matrix. A lower bound e^{-60n} for its smallest eigenvalue and an upper bound e^{50n} for the normalized matrix slope give stability on a neighborhood of radius e^{-120n} . The arithmetic-content and prime-reduction estimates of the construction then turn this local stability into a denominator bound. The final statements and their dependencies are verified in Lean. The exponent is explicit but not optimized, and no numerical denominator threshold is computed.

Keywords: irrationality exponent; zeta values; Diophantine approximation; Gram matrices; determinant estimates; formal verification.

1 Introduction

For an irrational real number x , its irrationality exponent is

$$\mu(x) = \sup\{\nu > 0 : 0 < |x - a/q| < q^{-\nu} \text{ for infinitely many reduced } a/q \in \mathbb{Q}, q \geq 2\}.$$

The supremum is taken in the extended reals. Dirichlet's theorem gives $\mu(x) \geq 2$. We use the conventional values $\zeta(s) = \sum_{m=1}^{\infty} m^{-s}$ for $s = 2, 3$.

Theorem 1.1. *For every $r \in \mathbb{Q}$, there is an integer $q_0(r) \geq 2$ such that*

$$\left| \zeta(3) - r\zeta(2) - \frac{a}{q} \right| > q^{-10000} \quad (q \geq q_0(r), a \in \mathbb{Z}, \gcd(a, q) = 1). \quad (1.1)$$

Consequently

$$2 \leq \mu(\zeta(3) - r\zeta(2)) \leq 10000.$$

The parameter r is fixed before the denominator tends to infinity. The exponent does not depend on r , but no uniform bound for $q_0(r)$ as the height of r grows is asserted. In particular, the theorem is not a simultaneous linear-independence measure with arbitrarily varying coefficients of $1, \zeta(2), \zeta(3)$.

The irrationality of $\zeta(3)$ is due to Apéry [1]. For $r = 0$, substantially stronger finite irrationality-exponent bounds are known; see Rhin and Viola [2]. The present extension concerns the full family with fixed rational r . Dauguet and Zudilin [3] constructed simultaneous approximations to $\zeta(2)$ and $\zeta(3)$ and lower bounds for linear forms with specific arithmetic constraints. Their approximation problem is distinct from the unrestricted rational approximants in (1.1).

Our starting point is Tang's proof [5] that $\zeta(3) - r\zeta(2)$ is irrational for every rational r , using Fauzan's determinant method [4]. We retain Tang's rational matrix pencil, primitive integer polynomials, arithmetic-content estimate, and nonvanishing after reduction modulo a prime. The analytic argument also uses his pointwise and logarithmic-energy estimates. These inputs are stated explicitly in Sections 2 and 3.

The additional estimates concern a positive Gram matrix associated with the signed moment matrix. We retain the positive multiple integral in the energy calculation and prove an exponential lower bound for its smallest eigenvalue. A separate residue estimate bounds the normalized slope. A bilinear determinant comparison then controls the entire pencil near the target number, without estimates for complementary minors. Finally, a prime in a logarithmic interval supplies nonvanishing at each rational approximant and yields (1.1).

Section 4 proves the conditioning estimate, Section 5 establishes local decay, and Section 6 proves Theorem 1.1. Section 7 records the formal statements and reproducible checks. The Lean source package includes the required upstream sources at fixed revisions.

2 The rational matrix pencil

Fix $r \in \mathbb{Q}$. For $n \geq 1$, put

$$\theta = \zeta(3) - r\zeta(2), \quad h = 3n, \quad D_m(t) = \prod_{j=1}^m (t + j), \quad R_n(t) = \frac{D_n(t)^4}{D_{5n}(t)},$$

and define

$$S_n = \frac{(5n)!}{(n!)^4}, \quad F_n = \prod_{j=0}^{h-1} (j!)^2, \quad t(y) = \frac{1}{2} + iy,$$

$$w_r(y) = \frac{\pi}{2} \operatorname{sech}^2(\pi y) (2r - 2\pi i \tanh(\pi y)), \quad \phi_n(y) = t(y) R_n(t(y)) w_r(y).$$

All matrix indices below run from 0 to $h - 1$.

We give the rational entries explicitly to fix their normalization. For $k \geq 0$, write

$$t^k R_n(t) = U_{n,k}(t) + \sum_{a=1}^{5n} \frac{c_{n,k,a}}{t+a}, \quad c_{n,k,a} = \frac{(-a)^k D_n(-a)^4}{\prod_{\substack{1 \leq b \leq 5n \\ b \neq a}} (b-a)}, \quad (2.1)$$

where $U_{n,k} \in \mathbb{Q}[t]$ is the polynomial part. Canceled poles have residue zero. Let $H_a^{(s)} = \sum_{b=1}^a b^{-s}$ and let B_e^+ be the Bernoulli numbers defined by

$$\frac{ze^z}{e^z - 1} = \sum_{e=0}^{\infty} B_e^+ \frac{z^e}{e!}; \quad B_1^+ = \frac{1}{2}.$$

Define a rational linear functional and rational coefficients by

$$\mathcal{L}_r(t^e) = (e+1)B_e^+ + 2rB_{e+1}^+, \quad b_{r,a} = 2r - 2aH_a^{(3)} + 2raH_a^{(2)}.$$

Tang's matrices are

$$(A_{r,n})_{jk} = \mathcal{L}_r(U_{n,j+k}) + \sum_{a=1}^{5n} c_{n,j+k,a} b_{r,a},$$

$$(B_n)_{jk} = \sum_{a=1}^{5n} 2a c_{n,j+k,a}. \quad (2.2)$$

The polynomial and resolvent moment identities of [5] are

$$\int_{\mathbb{R}} t(y)^{e+1} w_r(y) dy = (e+1)B_e^+ + 2rB_{e+1}^+, \quad \int_{\mathbb{R}} \frac{t(y)}{t(y)+a} w_r(y) dy = b_{r,a} + 2a\theta.$$

Applying them to (2.1) gives

$$(A_{r,n} + \theta B_n)_{jk} = \int_{\mathbb{R}} t(y)^{j+k} \phi_n(y) dy. \quad (2.3)$$

Define

$$Q_n(X) = \det(A_{r,n} + XB_n), \quad \tilde{Q}_n(X) = \frac{S_n^h}{F_n} Q_n(X).$$

When $Q_n \neq 0$, let $P_n \in \mathbb{Z}[X]$ be its primitive integer multiple with positive proportionality factor. Thus the greatest common divisor of the coefficients of P_n is one. Write $P_n = d_n \tilde{Q}_n$ with $d_n \in \mathbb{Q}_{>0}$. When $Q_n = 0$, set $P_n = 0$. The determinant expansion gives $\deg P_n \leq 3n$ whenever $P_n \neq 0$.

Proposition 2.1 (Arithmetic content and prime reduction, Tang). *For every $\varepsilon > 0$, all sufficiently large n with $Q_n \neq 0$ satisfy*

$$\log d_n \leq (283/50 + \varepsilon)n^2. \quad (2.4)$$

There is a finite exceptional prime set E such that every prime $p \geq 7$ outside E with $p \nmid \text{den}(r)$ satisfies

$$P_{p-1}(X) \pmod{p} \text{ is a nonzero constant.} \quad (2.5)$$

These are the arithmetic-content and prime-edge theorems of [5], for the definitions above. In particular, (2.5) implies $P_{p-1} \neq 0$, hence $Q_{p-1} \neq 0$. Our d_n is the source's **dtilde**: it is the primitive multiplier of the normalized polynomial. It is not the source's multiplier **d** of Q_n . The distinction accounts for the factor S_n^h/F_n in the analytic estimates.

3 The positive Gram determinant and its energy bound

Set

$$f_j(y) = \frac{t(y)^j}{j!}, \quad \rho_n(y) = S_n |\phi_n(y)|, \quad M_n = \left(\int_{\mathbb{R}} \overline{f_j(y)} f_k(y) \rho_n(y) dy \right)_{j,k < h}.$$

Every polynomial moment is integrable: the rational factor has no poles on $t = 1/2 + iy$, grows at most polynomially, and is multiplied by an exponentially decreasing hyperbolic kernel. The density is positive except possibly at $y = 0$. For $v \in \mathbb{C}^h$,

$$v^* M_n v = \int_{\mathbb{R}} \left| \sum_{j < h} v_j f_j(y) \right|^2 \rho_n(y) dy. \quad (3.1)$$

If $v \neq 0$, the polynomial inside the absolute value cannot vanish on an interval. Hence M_n is Hermitian positive definite. We write v^* for conjugate transpose and $\|v\|_2^2 = \sum_j |v_j|^2$.

Andréief's identity, followed by the Vandermonde determinant formula, gives

$$\det M_n = \frac{S_n^h}{F_n h!} \int_{\mathbb{R}^h} \prod_{\ell=1}^h |\phi_n(y_\ell)| \prod_{\ell < m} (y_\ell - y_m)^2 dy. \quad (3.2)$$

Indeed the two determinants of the functions f_j give a squared absolute Vandermonde divided by $\prod_j (j!)^2 = F_n$; the factors of i disappear in its absolute square. Absolute integrability permits the finite determinant expansion and Fubini's theorem.

Here is the precise positive quantity to which the energy estimates apply. For $u > 0$ set

$$W(u) = \frac{2}{3} \left[\pi u - 2 \left(u \arctan \frac{1}{u} + \frac{1}{2} \log(1 + u^2) \right) + \frac{1}{2} \left(u \arctan \frac{5}{u} + \frac{5}{2} \log(1 + u^2/25) \right) \right],$$

with $W(0) = 0$. Put

$$G_n(x) = \prod_{\ell=1}^h (1 + |x_\ell|)^7 e^{-3nW(|x_\ell|)} \prod_{\ell < m} (x_\ell - x_m)^2.$$

Lemma 3.1 (Pointwise and configuration estimates, Tang). *There are constants c_r, C , independent of n , such that, for $n \geq 1$,*

$$\rho_n(nx) \leq e^{c_r + 7 \log(n+1)} (1 + |x|)^7 e^{-3nW(|x|)}, \quad (3.3)$$

$$\int_{\mathbb{R}^{3n}} G_n(x) dx \leq \exp\left([-6 - 9(3/2 - \log 3)]n^2 + Cn(\log(n+1) + 1)\right). \quad (3.4)$$

The integrand in (3.4) is integrable.

Proof. We use the pointwise and configuration estimates in Tang's energy construction [5]. For clarity, the latter has the following form before its constants are combined. Its comparison probability density on a compact interval has logarithmic energy I and equilibrium constant ℓ , with

$$9(3/2 - \log 3) + 9(\ell - I) \leq -6.$$

The configuration bound supplies constants B, C_2 and the majorant

$$G_n(x) \leq e^{9n^2(\ell-I) + C_2 n(\log(n+1) + 1)} \prod_{j=1}^{3n} (1 + |x_j|)^7 e^{-3n \max(0, |x_j| - B)}.$$

Each one-variable factor has integral at most the fixed finite number

$$K = \max \left(1, \int_{\mathbb{R}} (1 + |x|)^7 e^{-\max(0, |x| - B)} dx \right).$$

Integrating, using the energy inequality, and absorbing $3n \log K$ into the error proves (3.4). The source results are `pointwise_bound`, `config_bound`, `IA_eq`, and the verified `FstarInput` certificate. These estimates concern a nonnegative integrand, not merely the absolute value of a signed determinant. $\square$

Proposition 3.2 (Positive energy bound). *For fixed r and every $\varepsilon > 0$, all sufficiently large n satisfy*

$$\det M_n \leq e^{(-6+\varepsilon)n^2}.$$

Proof. In (3.2), substitute $y_j = nx_j$. The Jacobian contributes n^h , and the squared Vandermonde contributes $n^{h(h-1)}$. Equation (3.3) therefore gives

$$\det M_n \leq \frac{n^{h^2} e^{h(c_r + 7 \log(n+1))}}{F_n h!} \int_{\mathbb{R}^h} G_n(x) dx.$$

The factorial sum, or Stirling's estimate summed over $j < h$, gives

$$\log F_n = h^2 \log h - \frac{3}{2} h^2 + O(h \log(h+1)).$$

With $h = 3n$ and $1/h! \leq 1$, the logarithm of the prefactor is at most $9(3/2 - \log 3)n^2 + O_r(n \log(n+1))$. Combining this with (3.4) proves

$$\det M_n \leq e^{-6n^2 + O_r(n \log(n+1))}.$$

The error divided by n^2 tends to zero, giving the assertion. $\square$

4 Exponential conditioning of the Gram matrix

Lemma 4.1 (Weight on a distant interval). *For $1 \leq x \leq 2$ and every $n \geq 1$,*

$$n\rho_n(nx) \geq e^{-22n},$$

uniformly in $r \in \mathbb{Q}$.

Proof. If $n \leq y \leq 2n$, each numerator factor of $R_n(t(y))$ has modulus at least n , and each denominator factor has modulus at most $8n$. Using $(5n)! \geq (5n/e)^{5n}$ and $n! \leq n^n$ gives

$$S_n |R_n(t(y))| \geq \left(\frac{5}{8e}\right)^{5n} \geq e^{-8n};$$

the last step follows from $\log(8/5) \leq 3/5$. Also $|t(y)| \geq n$ and $\tanh(\pi y) \geq 1/2$. The imaginary part of $2r - 2\pi i \tanh(\pi y)$ alone yields

$$|w_r(y)| \geq \frac{\pi^2}{2} e^{-2\pi y} \geq e^{-4\pi n},$$

since $\cosh u \leq e^u$ for $u \geq 0$ and $\pi^2/2 > 1$. Now $n^2 \geq 1$ and $8 + 4\pi < 22$ prove the claim. $\square$

Lemma 4.2 (Coefficient recovery). *Let $v \in \mathbb{C}^h$, where $h = 3n$ and $n \geq 1$, and put*

$$p(s) = \sum_{j < h} \frac{v_j}{j!} \left(\frac{1}{2} + in \frac{3+s}{2} \right)^j.$$

Then

$$\sum_{j < h} |v_j|^2 \leq 64h^5 111132^h \int_{-1}^1 |p(s)|^2 ds. \quad (4.1)$$

Proof. Write $M = \max_{[-1,1]} |p|$. We first recall the coefficient consequence of the endpoint Chebyshev derivative inequality. If a real polynomial g has degree at most h and $|g| \leq M$ on $[-1, 1]$, then

$$|g^{(j)}(1)| \leq MT_h^{(j)}(1), \quad |[s^i]g| \leq MT_h(3) \leq 7^h M.$$

For the middle inequality, expand g at 1, bound the coefficients of $(s-1)^j$ by 2^j , and sum the nonnegative Taylor coefficients of $T_h(1+z)$ at $z=2$. The recurrence $T_{h+2}(3) = 6T_{h+1}(3) - T_h(3)$ gives the last bound by induction. The derivative inequality is the standard extremal property of Chebyshev polynomials, also used in the formal proof. Apply this argument to the real and imaginary coefficient polynomials of p . Each has absolute value at most M on the interval, so

$$|[s^j]p| \leq 2 \cdot 7^h M.$$

For $s, t \in [-1, 1]$, the identity for $s^j - t^j$ gives $|s^j - t^j| \leq j|s - t|$. Consequently

$$|p(s) - p(t)| \leq 2h^2 7^h M |s - t|.$$

At a point where $|p| = M$, at least one direction has distance at least 1 to the endpoint of $[-1, 1]$. Along that direction take an interval of length $1/(4h^2 7^h)$. On this interval $|p| \geq M/2$, whence

$$M^2 \leq 16h^2 7^h \int_{-1}^1 |p(s)|^2 ds. \quad (4.2)$$

If $M = 0$, the conclusion is immediate. Otherwise define

$$q(z) = \sum_{j < h} \frac{v_j n^j}{j!} z^j = p(-3 + i/n - 2iz).$$

The affine constant has modulus at most 4 and the slope has modulus 2. Every coefficient of its j th power has modulus at most 6^j . Summing over the at most h coefficients of p yields $|[z^j]q| \leq 2h42^h M$. Since $j!/n^j \leq 3^h$ for $j < h = 3n$, we obtain $|v_j| \leq 2h126^h M$. Squaring and summing, then applying (4.2), proves (4.1), because $126^2 \cdot 7 = 111132$. $\square$

Proposition 4.3 (Gram lower bound). *There is N independent of r such that, for all $n \geq N$,*

$$v^* M_n v \geq e^{-60n} \|v\|_2^2 \quad (v \in \mathbb{C}^{3n}). \quad (4.3)$$

Equivalently, $\lambda_{\min}(M_n) \geq e^{-60n}$.

Proof. Restrict (3.1) to $y \in [n, 2n]$ and substitute $y = n(3 + s)/2$. Lemmas 4.1 and 4.2 give

$$v^* M_n v \geq \frac{e^{-22n}}{2} \int_{-1}^1 |p(s)|^2 ds \geq \frac{e^{-22n}}{128h^5 111132^h} \|v\|_2^2.$$

Since $111132 < e^{12}$ and $\log n = o(n)$, $128(3n)^5 111132^{3n} \leq e^{38n}$ for all sufficiently large n . Neither this threshold nor the preceding constants depend on r . $\square$

5 Slope growth and local stability

Normalize the whole pencil, for real X , by

$$H_n(X)_{jk} = \frac{S_n}{j!k!} (A_{r,n} + X B_n)_{jk}, \quad T_n = H'_n(X).$$

The row and column factorials give the identity

$$\det H_n(X) = \tilde{Q}_n(X). \quad (5.1)$$

It holds for every X , not only for $X = \theta$.

Lemma 5.1 (Normalized slope). *For $n \geq 1$ and $0 \leq j, k < 3n$,*

$$|(T_n)_{jk}| \leq e^{50n}.$$

Proof. Formula (2.1) gives, for $1 \leq a \leq 5n$,

$$S_n c_{n,0,a} = (-1)^{a-1} 5n \binom{5n-1}{a-1} \binom{a-1}{n}^4.$$

This also holds for $a \leq n$, when both sides vanish. The elementary bound $\binom{u}{v} \leq 2^u$ yields $|S_n c_{n,0,a}| \leq 5n 2^{25n}$. Since $c_{n,j+k,a} = (-a)^{j+k} c_{n,0,a}$, (2.2) implies

$$|(T_n)_{jk}| \leq 250n^3 2^{25n} \frac{(5n)^j}{j!} \frac{(5n)^k}{k!}.$$

Each factorial ratio is at most e^{5n} , by the exponential series. Also $250 \leq e^{8n}$, $n^3 \leq e^{3n}$, and $2^{25n} \leq e^{25n}$. Thus the displayed bound is at most $e^{46n} \leq e^{50n}$. $\square$

Lemma 5.2 (Bilinear Gram comparison). *Let M be a positive definite Hermitian $h \times h$ matrix and B a complex $h \times h$ matrix. If $C \geq 0$ and*

$$|u^T B v| \leq C \sqrt{u^* M u} \sqrt{v^* M v} \quad (u, v \in \mathbb{C}^h),$$

then $|\det B| \leq \det M C^h$.

Proof. Choose an invertible W with $W^*MW = I$, and set $L = W^\top BW$. Applying the hypothesis to Wu, Wv gives $|u^\top Lv| \leq C\|u\|_2\|v\|_2$. Take $u = \overline{Lv}$ to obtain $\|Lv\|_2^2 \leq C\|Lv\|_2\|v\|_2$, so $\|L\|_{2 \rightarrow 2} \leq C$. Hadamard's inequality gives $|\det L| \leq C^h$. Taking determinants in $W^*MW = I$ yields $|\det W|^2 \det M = 1$, while $|\det L| = |\det W|^2 |\det B|$. The result follows. The transpose in $W^\top BW$ is intentional: the form being estimated is bilinear, not sesquilinear. $\square$

For $F_v(y) = \sum_j v_j f_j(y)$, equation (2.3) gives

$$u^\top H_n(\theta)v = \int_{\mathbb{R}} F_u(y)F_v(y)S_n\phi_n(y) dy.$$

Since $|S_n\phi_n| = \rho_n$, Cauchy–Schwarz and (3.1) imply

$$|u^\top H_n(\theta)v| \leq \sqrt{u^*M_n u} \sqrt{v^*M_n v}. \quad (5.2)$$

All square integrals are finite by the moment integrability established in Section 3.

If E is a matrix with $|E_{jk}| \leq K$, finite Cauchy–Schwarz gives

$$|u^\top E v| \leq K \left(\sum_j |u_j| \right) \left(\sum_k |v_k| \right) \leq hK \|u\|_2 \|v\|_2.$$

For n satisfying Proposition 4.3, this is at most $hK e^{60n} \sqrt{u^*M_n u} \sqrt{v^*M_n v}$. Combine this with (5.2) and apply Lemma 5.2 to obtain

$$|\det(H_n(\theta) + E)| \leq \det M_n (1 + hK e^{60n})^h. \quad (5.3)$$

Proposition 5.3 (Local decay). *For fixed r , all sufficiently large n with $Q_n \neq 0$ satisfy*

$$|x - \theta| \leq e^{-120n} \implies |P_n(x)| \leq e^{-3n^2/10} \quad (x \in \mathbb{R}). \quad (5.4)$$

Proof. Use $\varepsilon = 1/100$ in Proposition 3.2 and (2.4). Set $E = H_n(x) - H_n(\theta)$ in (5.3). Lemma 5.1 permits $K = |x - \theta|e^{50n} \leq e^{-70n}$. Hence

$$|\tilde{Q}_n(x)| \leq e^{-599n^2/100} (1 + 3ne^{-10n})^{3n} \leq e^{-599n^2/100 + 3n}.$$

The last step uses $3n \leq e^{10n}$ and $2^{3n} \leq e^{3n}$. Multiplying by $d_n \leq e^{567n^2/100}$ gives

$$|P_n(x)| \leq e^{-8n^2/25 + 3n}.$$

For $n \geq 150$, the exponent is at most $-3n^2/10$. $\square$

6 Prime selection and the denominator bound

Lemma 6.1 (A prime avoiding a denominator). *For every fixed cutoff P_0 , all sufficiently large positive integers q admit a prime $p > P_0$ such that*

$$20 \log q < p \leq 60 \log q, \quad p \nmid q.$$

Proof. Write $\vartheta(u) = \sum_{p \leq u} \log p$. Elementary Chebyshev estimates give

$$\vartheta(u) \leq 2(\log 2)u, \quad \vartheta(u) \geq (\log 2)u - O(\sqrt{u} \log(u+2)).$$

Consequently $\vartheta(3u) - \vartheta(u) \geq (\log 2)u - o(u) > u/4$ for sufficiently large u . Take $u = 20 \log q$. If every prime in $(u, 3u]$ divided q , their product would divide q , so the sum of their logarithms would be at most $\log q$. Its lower bound $5 \log q$ contradicts this. Finally $u \rightarrow \infty$ places the window above P_0 . Only elementary prime estimates are needed for this selection step. $\square$

Proof of Theorem 1.1. Fix r . Choose P_0 above 7, every exceptional prime in (2.5), $\text{den}(r)$, and all fixed thresholds needed for Proposition 5.3, increased by one. For sufficiently large q , take p from Lemma 6.1 and put $n = p - 1$. Increasing the threshold for q if necessary gives

$$19 \log q < n \leq 60 \log q. \quad (6.1)$$

Since $p \nmid \text{den}(r)$, (2.5) applies and gives $Q_n \neq 0$. Suppose that a reduced rational a/q satisfies $|\theta - a/q| \leq q^{-10000}$. By (6.1),

$$q^{-10000} = e^{-10000 \log q} \leq e^{-120n},$$

because $120 \cdot 60 = 7200 < 10000$. Proposition 5.3 gives

$$|P_n(a/q)| \leq e^{-3n^2/10} < q^{-3n}; \quad (6.2)$$

the strict inequality uses $n > 10 \log q$.

On the other hand, reduction modulo p evaluates P_n to a nonzero constant. The denominator q is invertible modulo p , so $P_n(a/q)$ cannot vanish. Since $P_n \in \mathbb{Z}[X]$ has degree at most $3n$,

$$q^{3n} P_n(a/q) \in \mathbb{Z} \setminus \{0\}.$$

Thus $|P_n(a/q)| \geq q^{-3n}$, contradicting (6.2). This proves (1.1) for every sufficiently large reduced denominator, not merely for a selected sequence.

Tang's theorem supplies the irrationality of θ , so Dirichlet's theorem gives $\mu(\theta) \geq 2$. For $\nu \geq 10000$, (1.1) excludes all sufficiently large reduced denominators from $0 < |\theta - a/q| < q^{-\nu}$. For the finitely many remaining denominators only finitely many numerators can satisfy this inequality. The defining set of approximation exponents is therefore bounded above by 10000, proving the upper bound. $\square$

The numerical choices have substantial slack: local stability requires only the comparison with 7200 above, and the integer-polynomial contradiction needs $n > 10 \log q$, whereas the prime window gives $n > 19 \log q$. We keep 10000 as the stated and formally checked exponent. Computing $q_0(r)$ or optimizing these constants is a separate problem.

7 Formal verification and reproducibility

The two final declarations, in the namespace `Zeta32Extension`, are

```
denominator_bound,    irrationalityExponent_bounds.
```

The first asserts, for every $r : \mathbb{Q}$, that eventually in natural b , every rational z with reduced denominator b satisfies

$$b^{-10000} < |\text{Cr}(r) - z|, \quad \text{Cr}(r) = \sum_{k=0}^{\infty} \frac{1}{(k+1)^3} - r \sum_{k=0}^{\infty} \frac{1}{(k+1)^2}.$$

Thus the formal target is the literal series expression in the theorem. There is no analytic, arithmetic, matrix-rank, or nonvanishing hypothesis left in either endpoint. The exponent is defined as a real supremum; before using it the code proves that its defining set contains 2 and is bounded above. It therefore agrees here with the extended-real definition in Section 1.

The following correspondence identifies the principal modules. Extension paths are relative to `Zeta32Extension/`; Tang's modules are relative to `Zeta32/`.

Mathematical component	Principal source modules
Pencil, primitive normalization, and Proposition 2.1	<code>Family.lean</code> , <code>Final.lean</code> (Tang)
Positive integral and Proposition 3.2	<code>PositiveEnergy.lean</code> , <code>PositiveEndpoint.lean</code> , <code>GramDeterminant.lean</code>
Weight and coefficient recovery	<code>WeightLower.lean</code> , <code>RealPolynomialCoefficient.lean</code> , <code>ComplexPolynomialCoefficient.lean</code> , <code>PolynomialIntegralLower.lean</code> , <code>PolynomialConditioning.lean</code>
Proposition 4.3	<code>ConditioningCost.lean</code> , <code>WeightedQuadratic.lean</code> , <code>GramQuadratic.lean</code>
Slope and normalized pencil	<code>SlopeGrowth.lean</code> , <code>Pencil.lean</code>
Bilinear comparison and local decay	<code>SignedBilinear.lean</code> , <code>GramComparison.lean</code> , <code>BilinearPerturbation.lean</code> , <code>DirectStability.lean</code>
Prime selection and final exponent	<code>PrimeWindow.lean</code> , <code>ConditionalExponent.lean</code> , <code>UnconditionalExponent.lean</code> , <code>ExponentCorollary.lean</code>

The reusable denominator reduction takes local decay as a premise. The direct Gram comparison proves that premise, so the final denominator theorem is unconditional. A separate minor-based conditional theorem is not used to prove the final result. The elementary approximation-set lemmas and a Hilbert-space determinant lemma are adapted from the OpenAI formal developments [6], with attribution in the corresponding files. Neither their π nor their Catalan endpoint theorem is an assumption here.

The formalization fixes Lean v4.35.0-rc3, Mathlib revision

`c55e6e786f49471c72fbddbec5415808896aec1e`,

and Tang’s source revision

`669c92bf7a3728e0de9ad80c373bc29fb5ca3d58`.

The source archive contains the extension, the required unmodified upstream mathematical sources, pinned Lake configuration, licenses, and checksum inventories. From `anc/lean`, run sequentially

```
lake exe cache get
lake build Solution Zeta32Extension.Verification
lake env lean Zeta32Extension/Verification.lean
lake env lean -run VerifyZeta32Components.lean
```

The build and transitive axiom audit passed for all 123 listed extension theorems. A replay in an initially empty Lean-kernel environment checked 21 selected roots and their closure of 82674 declarations, including both endpoints. Only `propext`, `Classical.choice`, and `Quot.sound` were permitted; unsafe, partial, and nonstandard-axiom dependencies were rejected. The replay uses Lean’s own kernel. The checks reuse pinned dependency caches; the package does not contain compiled proof objects or claim a clean-machine rebuild of Mathlib. Do not rebuild imported modules while the replay is running. Further instructions are in `anc/README.txt`.

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